

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I Lokesh S (192511037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lokesh S

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

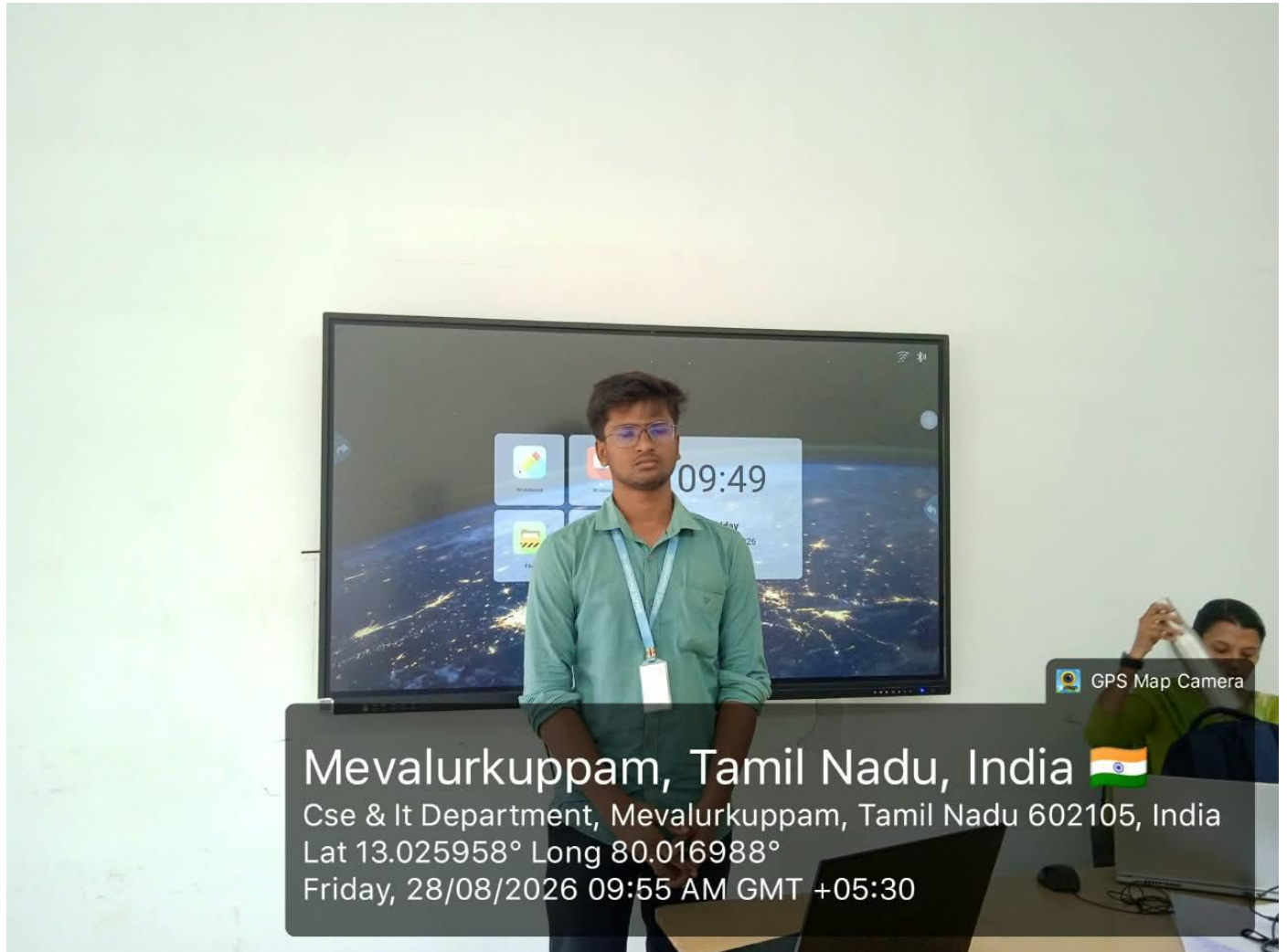
5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Multi-Agent System For Disaster Management



2. Need for a Multi-Agent System

- Disasters generate large amounts of data from sensors, cameras, satellites and people.
- Centralized systems may face processing delays and communication bottlenecks.
- Different locations require local decisions while maintaining overall coordination.
- Autonomous agents can detect abnormal conditions and react with low latency.
- Cooperation between agents enables faster detection, resource allocation and emergency response.

3. System Architecture

Sensors → Agents → Coordination → Decision → Emergency Response

Sensor Agents

Rainfall • Water level • Smoke • Temperature • Cameras

Data Collection

Filter, validate and timestamp observations

Communication Layer

Event bus / APIs / secure agent-to-agent messaging

Decision Agents

Risk assessment • data fusion • prediction

Coordination

Consensus • task allocation • priority handling

Response Agents

Alerts • evacuation • rescue • resources

- Sensor agents observe the environment and generate events.
- Decision agents combine evidence and calculate disaster risk.
- Communication and coordination agents synchronize the response.
- Response agents deliver alerts and operational actions.
- Feedback from new observations continuously updates decisions.

4. Types of Agents

- **Sensor Agents:** collect rainfall, water level, temperature, smoke, camera and GPS data.
- **Decision Agents:** analyze data, estimate risk and select appropriate response actions.
- **Communication Agents:** distribute warnings and operational information to stakeholders.
- **Coordination Agents:** assign tasks, resolve conflicts and maintain a common situation picture.
- **Response Agents:** support evacuation, rescue-team dispatch, resource allocation and public alerts.

5. Coordination and Data Sharing

- Publish/Subscribe allows agents to exchange only the events relevant to their tasks.
- Shared blackboards provide a common workspace for observations, alerts and decisions.
- Consensus mechanisms help agents agree when sensor observations conflict.
- Standard event formats can contain event ID, location, timestamp, severity and confidence.
- Authentication, authorization, encryption and audit logs protect disaster information.

6. Real-Time Disaster Response

- Sensor agents detect rising river levels and heavy rainfall in a vulnerable area.
- Decision agents fuse the observations and classify the situation as high flood risk.
- Coordination agents identify evacuation zones and assign nearby rescue resources.
- Communication agents send prioritized warnings to authorities, responders and citizens.
- New sensor data is continuously processed so the response plan can be updated in real time.